Artificial Intelligence Technology and Application

ModelArts Lab Guide

Student Version

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About This Document

Description

This document presents one exercise on Huawei Cloud ModelArts ExeML.

- Exercise 1: How to use Huawei Cloud ModelArts ExeML

Knowledge Required

This course is for Huawei's development certification. To better understand this course, familiarize yourself with:

- Basic computer operations.

Lab Environment

- Accessible network

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1 ExeML

1.1 Introduction

ExeML, a service provided by ModelArts, automates model design, parameter tuning and training, and model compression and deployment using labeled data. ExeML is free of coding and does not require you to have experience in model development, meaning that it is beginner friendly. The exercise covered in this course will help you understand and use image classification, object detection, and predictive analysis.

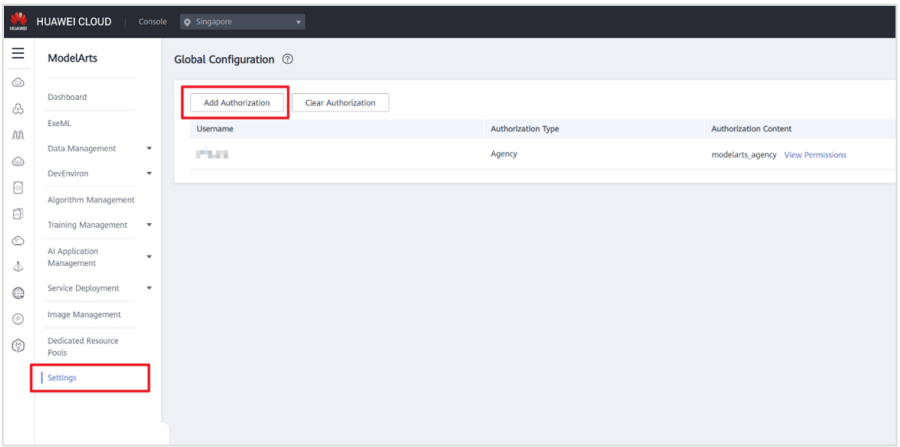
Image classification identifies a class of objects in images. An image classification model can predict the label of an image, and applies to scenarios in which image classes are distinguishable.

1.2 Objectives

The exercise in this course uses flower recognition as an example, to help you quickly create and learn about image classification models.

1.3 Lab Environment

If you are using ModelArts for the first time, authorize ModelArts to access Huawei Cloud Object Storage Service (OBS). Without doing so, you will not be able to create jobs on ModelArts. The procedure is as follows.



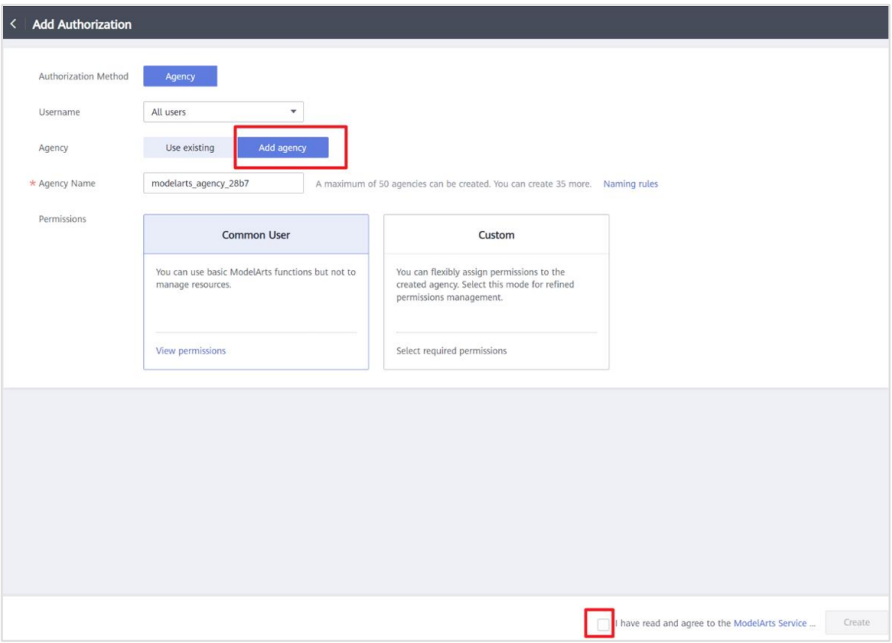


Figure 1-1 ModelArts console

1.4 Flower Recognition Exercise

The ModelArts ExeML page consists of two parts. The upper part lists the supported project types. You can click **Create Project** to create an ExeML project. The lower part displays the list of created ExeML projects. You can filter projects by project type in the upper right corner of the list, or enter key words in the text box and click to search for projects that contain those key words.

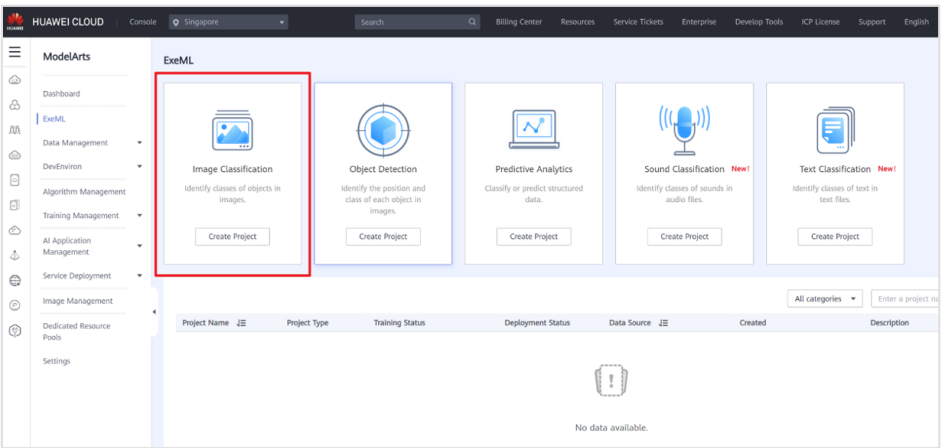


Figure 1-2 ExeML

The procedure of the exercise consists of four parts:

- Creating a project: To use ModelArts ExeML, first create an ExeML project.
- Labeling data: Upload images and label them by class.
- Training a model: After data labeling, train a model.
- Deploying a service and performing prediction: Deploy the model as a service and perform real-time prediction.

1.4.1 Creating a Project

Step 1 Set parameters.

On the **ExeML** page, click **Create Project** in **Image Classification**.

The **Create Image Classification Project** page is displayed.

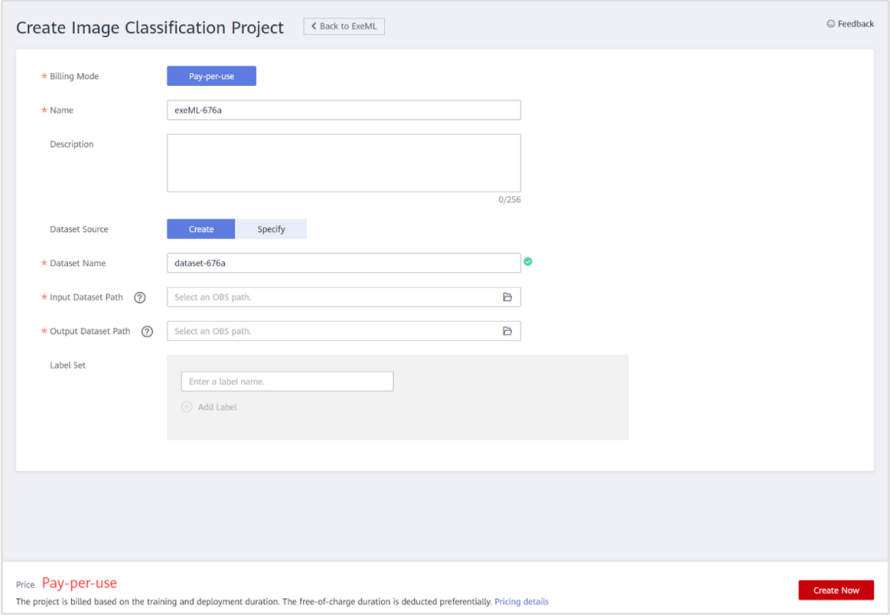


Figure 1-3 Creating a project

Parameters:

Billing Mode: **Pay-per-use** by default

Name: Enter a project name.

Dataset Source: Select the OBS path where the dataset is stored.

You must create an empty OBS folder and select this folder as the training data path. (Click the bucket name, and then click **Create Folder**.) Alternatively, import the data to OBS in advance. In this example, click the following data link and decompress the downloaded file package, and upload the data in the **train** folder to the **OBS/ExeML/flower/train** directory.

For details about how to upload data, see https://support.huaweicloud.com/intl/en-us/modelarts_faq/modelarts_05_0013.html.

Data link:

https://certification-data.obs.cn-north-4.myhuaweicloud.com/ENG/HCIA-AI/V3.5/chapter5/ExeML.zip

Step 2 Click **Create Now**.

Click **Create Now**. The ExeML project is then created.

1.4.2 Labeling Data

Step 1 Upload images.

After an ExeML project is created, the data labeling page is automatically displayed. Images uploaded to the OBS directory configured during project creation are automatically loaded. Labeled images are displayed in the **Labeled** tab.

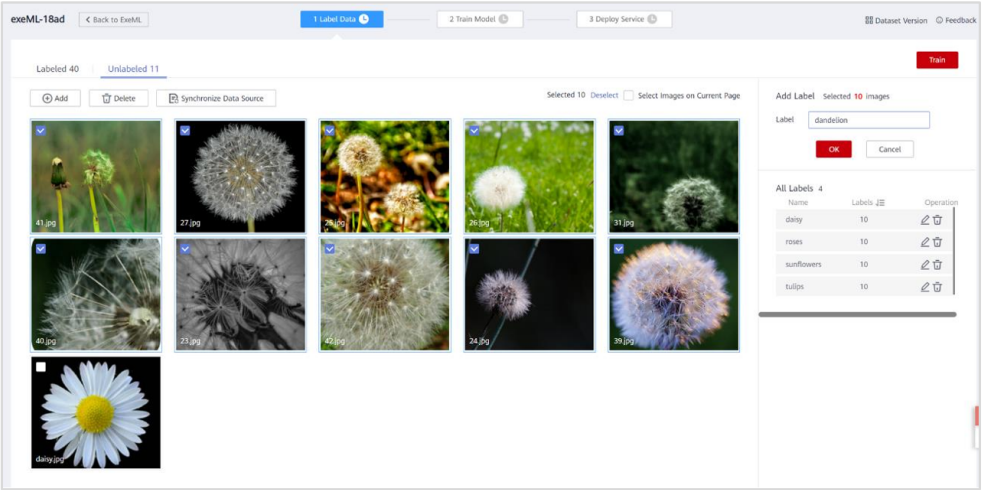


Figure 1-4 Data labeling for image classification

Notes:

- The images to be trained must be classified into at least two classes, and each class must contain at least five images. That is, at least two labels are available and the number of images for each label is not fewer than five. In this example, the dataset contains five labels: tulips, daisy, sunflowers, roses, and dandelion.
- You can add multiple labels to an image.

Step 2 Label the images.

Click **Unlabeled** and select the unlabeled images you want to label, or select **Select Images on Current Page** in the upper right corner to select all images on the page. Then, input a label or select an existing label from the drop-down list, and then press **Enter**. Then, click **OK**.

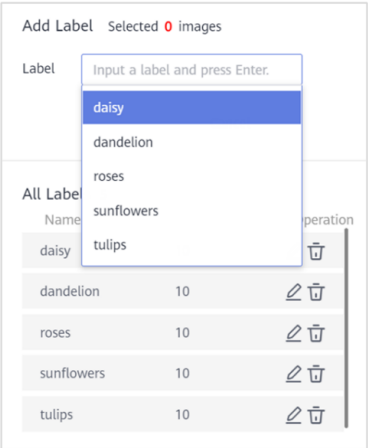


Figure 1-5 Image labeling for image classification

Step 3 Modify labels.

Click the **Labeled** tab, select an image, enter a new label name in the **Label** text box on the right, and click **OK**.

You can select multiple images at a time and modify them in batches.

1.4.3 Training a Model

After labeling the images, train a model. Before training, set parameters. Images to be trained must be classified into at least two classes, and each class must contain at least five images. Therefore, before training, ensure that the labeled images meet the requirements. Otherwise, the **Train** button remains unavailable.

Step 1 Set parameters.

Use default settings or configure them.

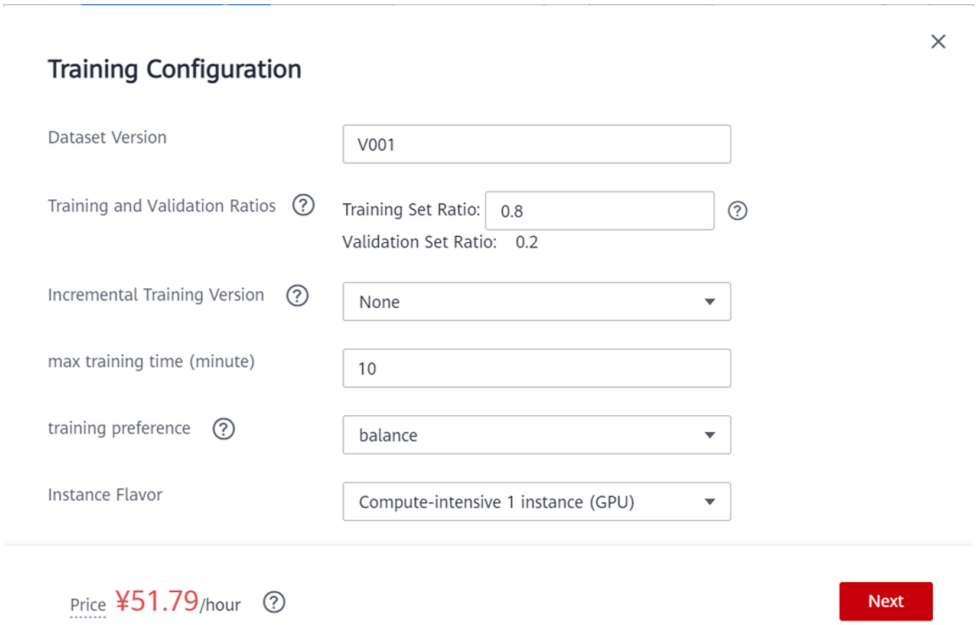


Figure 1-6 Training Configuration

Parameters:

max training time (minute): If the training is not completed within the maximum training duration, the training is forcibly stopped. Enter a larger value to avoid this situation.

Step 2 Train a model.

After setting, click **Train**. After the training, you can view the training result on the **Train Model** tab page.

1.4.4 Deploying a Service and Performing Prediction

Step 1 Deploy a service.

After model training, deploy a model version with satisfactory accuracy and in the **Completed** status as a service. To do so, click **Deploy** in **Version Manager** of the **Train Model** tab page. After the deployment, choose **Service Deployment** > **Real-Time Services** to view the deployed service.

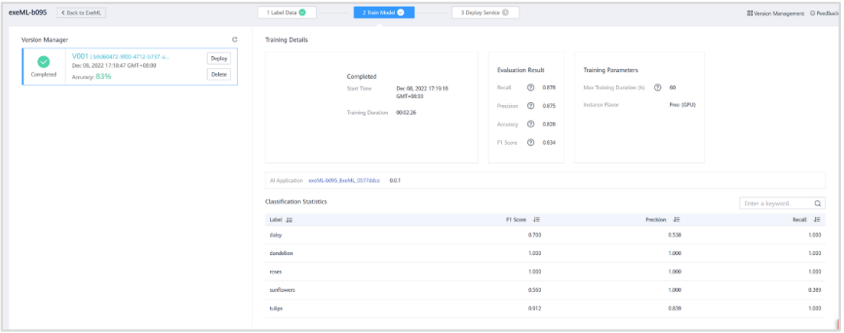


Figure 1-7 Deploy

Step 2 Test the service.

After the model is deployed, test the service using an image. The test data is stored in **OBS/flower/test/daisy.jpg**. On the **Deploy Service** tab, click **Upload** and upload an image. Then, click **Predict**. The test result is displayed in the right pane. There are five labels for data labeling: tulip, daisy, sunflower, rose, and dandelion. In the prediction, "daisy" gets the highest score, so the classification result is "daisy".

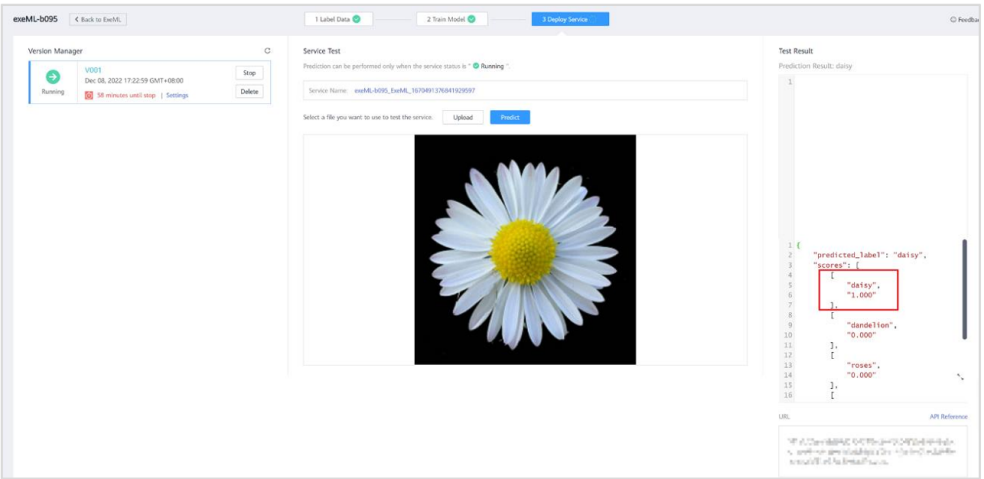


Figure 1-8 Service test